

The pretentious style

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I. L'INTRODUCTION

INDOUBETABLY, there shall come a time in all of our lives when taste and sophistication must reign supreme, when the much more idle forms of typography must give way to class and distinction. Ah yes, Garamond, what a beautiful, elegant font—oh, how its form reminds me of bright summer days long past in this cruel winter of *grotesque* sans serifs and transitional serif fonts. Unlike those other styles, this one is cosmopolitan and refined, urbane as they come—it has read *Ulysses* and Russell's *Principia Mathematica*, listened to Liszt, and wrote poetry with words like “ichor” and “vivacious,” and wrote the plot to *Deus Ex: Human Revolution*. Truly, no other style set would ever be quite as cultured as this.

1.1. Ok, I don't actually hate you if you like this style

In reality, I love old-styled, humanist typesetting, just like this. Aside from looking “sophisticated,” styles like this translate to print very nicely—provided your printer has sufficient DPI—and it does have a certain *je ne c'est quoi* about it, a certain style that is tantalizing to me. I did not mean to call this style pretentious to prevent you from using it—if this is your style, by all means use it—but the style does carry an element of ostentatiousness about it. The truth is, you *should* use this style.

There are many things which simply look good in Garamond; there are many times when you simply need something to be Refined and Cultured™. I can't think of a particular example right now. In any case, if you want to impress a general audience, you want to give off an air of sophistication, use this. Typesetting has rhetorical power, whether people know it or not. In that sense, this is the rhetorical equivalent of a champagne glass—something which practically *begs* for high-quality contents, but can make apple juice look fancy. (Please typeset responsibly.)

I would avoid using this style set if your audience has a lot of low resolution screens or your printer has a low DPI count. Garamond is very pretty, but the typeface demands high resolution to be readable. Likewise, you should not hand somebody a paper using this style while wearing a powdered wig and tights, primarily because you would look very silly while doing so.

2. STANDARD DEMO

3. STANDARD DEMO

The following text is taken from Lieb, E., Schultz, T., Mattis, D. (2004).¹ Please, go read the actual paper, it's great.

3.1. Formulation

The first model consists of N spin $1/2$'s (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i [(1 + \gamma) S_i^x S_{i+1}^x + (1 - \gamma) S_i^y S_{i+1}^y],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy -plane. Because the Hamiltonian only involves the x - and y - components of the spin operators, we call this model the XY model.

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + i S_i^y \text{ and } a_i = S_i^x - i S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = (a_i^\dagger + a_i)/2; \quad S_i^y = (a_i^\dagger - a_i)/2i; \quad S_i^z = a_i^\dagger a_i - \frac{1}{2}$$

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i [(a_i^\dagger a_{i+1} + \gamma a_i^\dagger a_{i+1}^\dagger) + \text{h.c.}].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k \Psi_0 = 0, \text{ all } k.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k.$$

¹Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) Condensed Matter Physics and Exactly Soluble Models. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

In the limit $N \rightarrow \infty$, the sum can be replaced by an integral giving

$$\begin{aligned}\frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} [1 - (1 - \gamma^2) \sin^2 k]^{1/2} dk \\ &= -\frac{1}{\pi} \mathcal{E}(1 - \gamma^2),\end{aligned}$$

where $\mathcal{E}(k^2)$ is one of the complete elliptic integrals (13).